

MOHAMED SHABAN

Senior Software Engineer | Backend / Full-Stack Engineer | Application Security

Giza, Egypt | +20 114 310 4499 | mohamedgmshaban@gmail.com | linkedin.com/in/mohamed-shaban-581ba9274 | github.com/mohamedgitshaban

PROFESSIONAL SUMMARY

Senior Software Engineer with 4+ years of experience building scalable backend and full-stack applications using PHP, Laravel, React, REST APIs, MySQL, and Redis, across ERP, insurance, payment, e-commerce, HR, logistics, and real-time systems. Hands-on application-security practitioner: independently investigated and remediated a full gray-box penetration test covering both the Tenant and Super Admin portals of an insurance platform, fixing multiple Critical and High findings spanning broken access control, business logic, session management, and rate limiting. Currently deepening expertise in Java, Spring Boot, distributed systems, system design, and cloud engineering (IBM-certified in Java).

TECHNICAL SKILLS

- **Backend:** PHP, Laravel, Java, Spring Boot, REST APIs, Queues, Webhooks, Authentication, Authorization
- **Frontend:** React.js, Next.js, JavaScript, HTML5, CSS3, Tailwind CSS, Bootstrap, Material UI
- **Database:** MySQL, SQL, Database Design, Query Optimization, Redis, Caching, Transactions
- **Architecture:** OOP, SOLID, Design Patterns, System Design, Distributed Systems Concepts, Microservices Concepts, Idempotency
- **Cloud/DevOps:** AWS S3, Docker, Linux, CI/CD Concepts, Object Storage, Deployment
- **Application Security:** OWASP Top 10, Broken Access Control & Privilege-Escalation Remediation, RBAC, API Security, JWT & Session Hardening, Rate-Limiting Design, Secrets Management, Secure Object Storage, Vulnerability Triage
- **Tools:** Git/GitHub, Composer, npm, PHPUnit, Fawry, BayMob, Google Routes, Firebase, Twilio

PROFESSIONAL EXPERIENCE

Senior PHP Developer — Dafa

Cairo, Egypt | Oct 2024 – Present

- Lead backend development using PHP, Laravel, MySQL, Redis, REST APIs, queues, and third-party integrations for insurance, payment, wallet, logistics, and digital-asset workflows.
- Served as the primary developer responding to an external gray-box penetration test of the Tenant and Super Admin portals; triaged and remediated 2 Critical, 2 High, 3 Medium and multiple Low/Informational findings, including:
- Closed a Critical broken-access-control gap that let low-privileged users view third-party payment-gateway (Paymob) credentials, by enforcing server-side, role-based authorization on settings endpoints and removing secrets from client-facing API responses.
- Fixed a High-severity self-role-modification flaw by adding server-side checks that block users from elevating their own role/permissions, closing a privilege-escalation path to Super Admin.
- Resolved a High-severity business-logic flaw in the insurance claims workflow that let negative claim amounts bypass annual coverage limits, by adding strict server-side validation on claim approval amounts.
- Secured a publicly-listable object storage bucket (Critical) by disabling directory listing and enforcing authenticated, time-limited access instead of static public URLs.
- Locked down unauthenticated API endpoints exposing user PII, claims/financial data, and an unauthenticated user-delete endpoint, by adding authentication and object-level authorization checks across the API.
- Redesigned rate-limiting to fix a shared-IP/proxy misconfiguration causing denial-of-service and an IP-rotation bypass, moving to per-account throttling on top of client-IP checks.
- Hardened session management: shortened JWT lifetimes, added HttpOnly/Secure/SameSite cookie attributes, and enforced session invalidation on authenticated password change.
- Removed the exposed default Laravel welcome page and PHP version disclosure, and upgraded vulnerable front-end dependencies (Quill Editor, Lodash).
- Manage AWS S3/object storage and Dockerized application environments; optimize database queries, indexing, caching, and asynchronous processing for production performance.

PHP Developer — STARON Egypt

Cairo, Egypt | Jul 2023 – Sep 2024

- Developed ERP, e-commerce, payment, and logistics systems using Laravel, PHP, MySQL, and React.
- Built dashboards, REST APIs, payment workflows, and real-time mobile synchronization.

Full Stack Developer — RSTAR

Cairo, Egypt | Jun 2022 – Jun 2023

- Developed Laravel backend systems and REST APIs with optimized MySQL database schemas.
- Built React interfaces and integrated frontend applications with backend services.

Software Instructor — ITSHARE

Cairo, Egypt | Dec 2021 – Jun 2022

- Delivered practical training in HTML, CSS, JavaScript, React, PHP, Laravel, and Spring Boot.
- Guided students in developing full-stack applications and applying OOP, databases, REST APIs, and debugging practices.

SELECTED PROJECTS

Wathiqaty — Insurance Platform: Application Security Remediation

Laravel | PHP | MySQL | REST APIs | RBAC | JWT | Rate Limiting

- Owned end-to-end remediation of an external gray-box penetration test covering the Tenant Dashboard and Super Admin Portal (insurance providers, tenants, admins, customers, policies, claims, payments, eKYC, documents, e-signatures).
- Fixed Critical/High findings in authorization, object storage, business logic, session security, and rate limiting; coordinated verification with the security vendor and tracked remediation status per finding.

Maqsafy — Electronic Payment Platform | Saudi Arabia

Laravel | MySQL | Redis | Laravel Nova | Payment Gateways

- Built wallet and financial workflows including transfers, deposits, withdrawals, commissions, refunds, reconciliation, school fees, and order management.
- Integrated Moyasar, HyperPay, Mastercard, Apple Pay, and MADA with secure payment handling.
- Implemented scoped RBAC, background jobs, Laravel Excel exports, Firebase notifications, AWS S3 storage, and database optimization.

Goway — Ride-Hailing Platform

Laravel 10 | Redis | Ably/Reverb | Google Routes | Payments

- Developed REST APIs and complete trip lifecycle workflows including matching, live tracking, pricing, cancellation, wallets, and payments.
- Implemented real-time driver discovery using geohashing, Redis, Ably/Reverb, and dynamic pricing using distance, traffic, tolls, coupons, and surge factors.

ROX Custody Integration

Laravel | REST APIs | Webhooks | Background Jobs

- Built secure REST API integration with ROX Custody using API-key authentication and reusable HTTP clients.
- Implemented wallet synchronization through webhooks, event logging, caching, optimized queries, and background jobs.

STARON Egypt ERP System

Laravel | React | MySQL | REST APIs

- Developed ERP and e-commerce functionality covering payments, logistics, dashboards, and mobile synchronization.

Agrigroup HR & Payroll System

Laravel | MySQL | REST APIs

- Built employee management, attendance, deductions/additions, vacation/warning logs, and monthly payroll generation.
- Implemented QR-based employee lookup, secure APIs, logging, and error handling.

EDUCATION

Bachelor of Computer Science and Information — Cairo University | Cairo, Egypt | Nov 2017 – Jan 2022

CERTIFICATIONS

- IBM Java Programming Certificate — IBM
- Mahara-Tech — Spring Boot
- Mathematics for Machine Learning: Linear Algebra — Imperial College London

LANGUAGES

Arabic: Native | English: Professional Working Proficiency